



THREATWATCH AI

**IS 3060-301 SPRING
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Business Introduction

• Problem Context

Alert Overload

- SOC teams face high volumes of mixed alerts, so truly critical signals can be missed.
- The 2025 SK Telecom incident shows how quickly a few high-severity signals can escalate into major business impact if not surfaced early.

Inconsistent Triage Decisions

- Severity labels (P1/P2/P3) can vary by analyst judgment, causing wrong prioritization.
- For telecom-scale assets (e.g., USIM/subscriber data), misclassification can delay escalation and increase regulatory exposure—as seen in the SKT case

Delayed Escalation

- Manual context gathering and summarization slows first response when speed matters most.
- SKT's 2025 breach highlights why rapid escalation for high-risk cases is essential.

Weak Auditability

- Without structured rationale + missing-data tracking, audits and post-incident learning are difficult.
- After incidents like SKT's, regulators and stakeholders expect clear decision trails and documented actions.



Business Introduction

- **Our Solution**

Goal: Turn noisy security alerts into consistent, auditable, and fast escalation decisions using an LLM + automation workflow.

Why this solution fits the Five Forces pressure:

- Rivalry (Industry competition): Security reliability is now a key differentiator → faster, more consistent triage protects uptime and brand trust.
- Buyer Power (Customer expectations): Customers can demand transparency and quick response → structured summaries + decision trails improve accountability.
- Threat of Substitutes (Switching options): If trust drops, users can switch to alternatives → rapid escalation reduces prolonged incidents that trigger churn.
- Supplier Power (Vendors/Platforms): Tool stacks are complex and vendor-driven → our design is modular (n8n orchestration + Python validation), reducing lock-in risk.
- Threat of New Entrants (AI-enabled competitors): AI lowers the cost of operational excellence → adopting LLM-assisted triage helps us keep pace with AI-driven security operations.

What ThreatWatch AI does (one-line workflow):

Alert input → LLM risk assessment (level/score/type/summary) → schema validation → route action (email/log) → Google Sheets audit log.



BPMN without LLM Integrated

Before AI(2020)

Porter's Five Forces:
Security-Critical Digital Service
Providers (Baseline Analysis,
Pre-AI)
Prepared by: Sejin Kim, Chaehoon
Lee
Date: Jan 26, 2026



Legend

- High Intensity (bad for incumbents)
- Medium Intensity
- Low Intensity (good for incumbents)

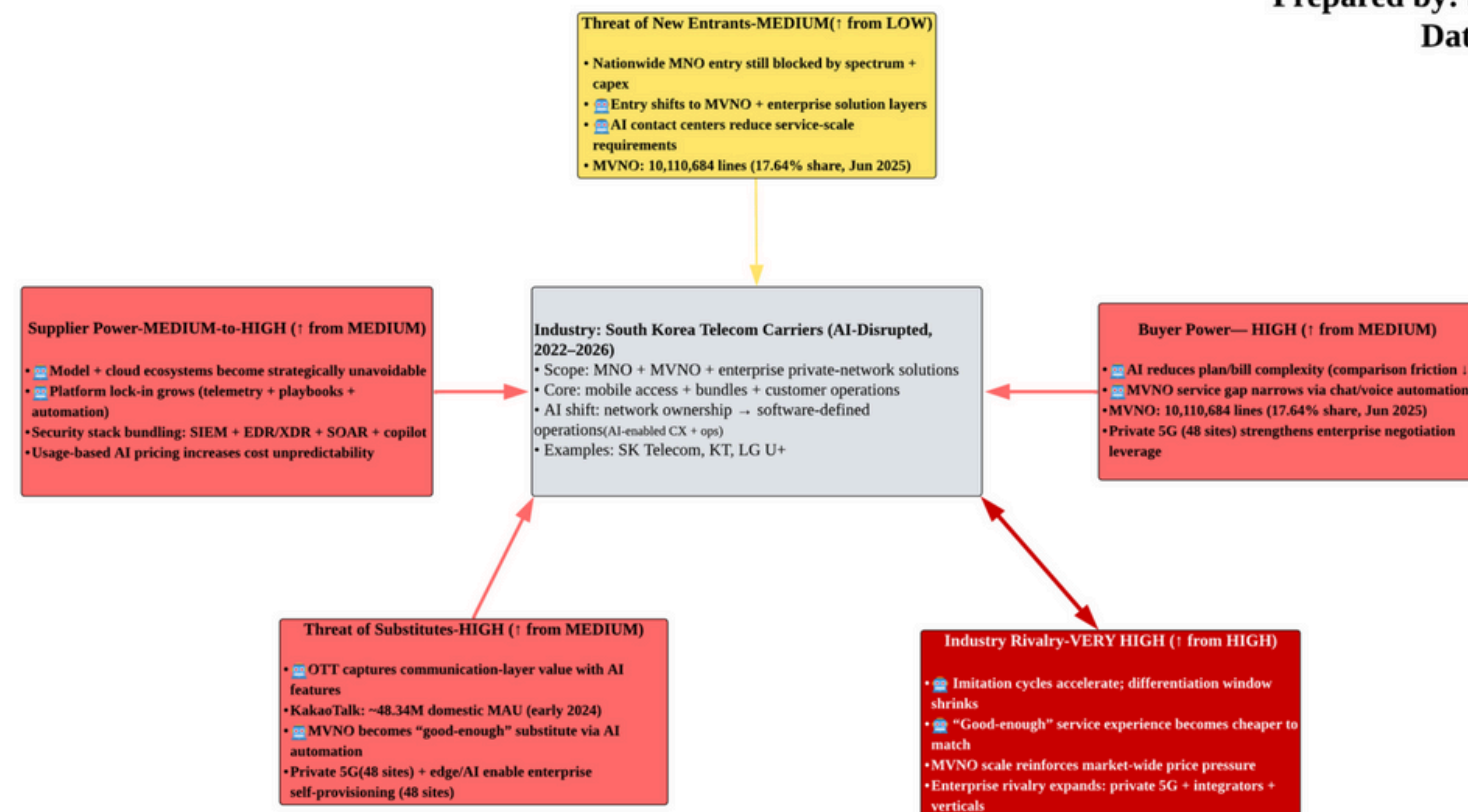
- High rivalry, trust is the differentiator
- Triage is manual → slow and inconsistent decisions
- Context gathering is fragmented across tools
- Result: human-driven workflow with high variance



BPMN with LLM Integrated

After AI(2026)

Porter's Five Forces: South Korea Telecom Carriers
(MNO + MVNO + Enterprise Private Networks)
AI Disruption Analysis (2024–2026)
Prepared by: Sejin Kim, Chaehoon Lee
Date: Jan 28, 2026

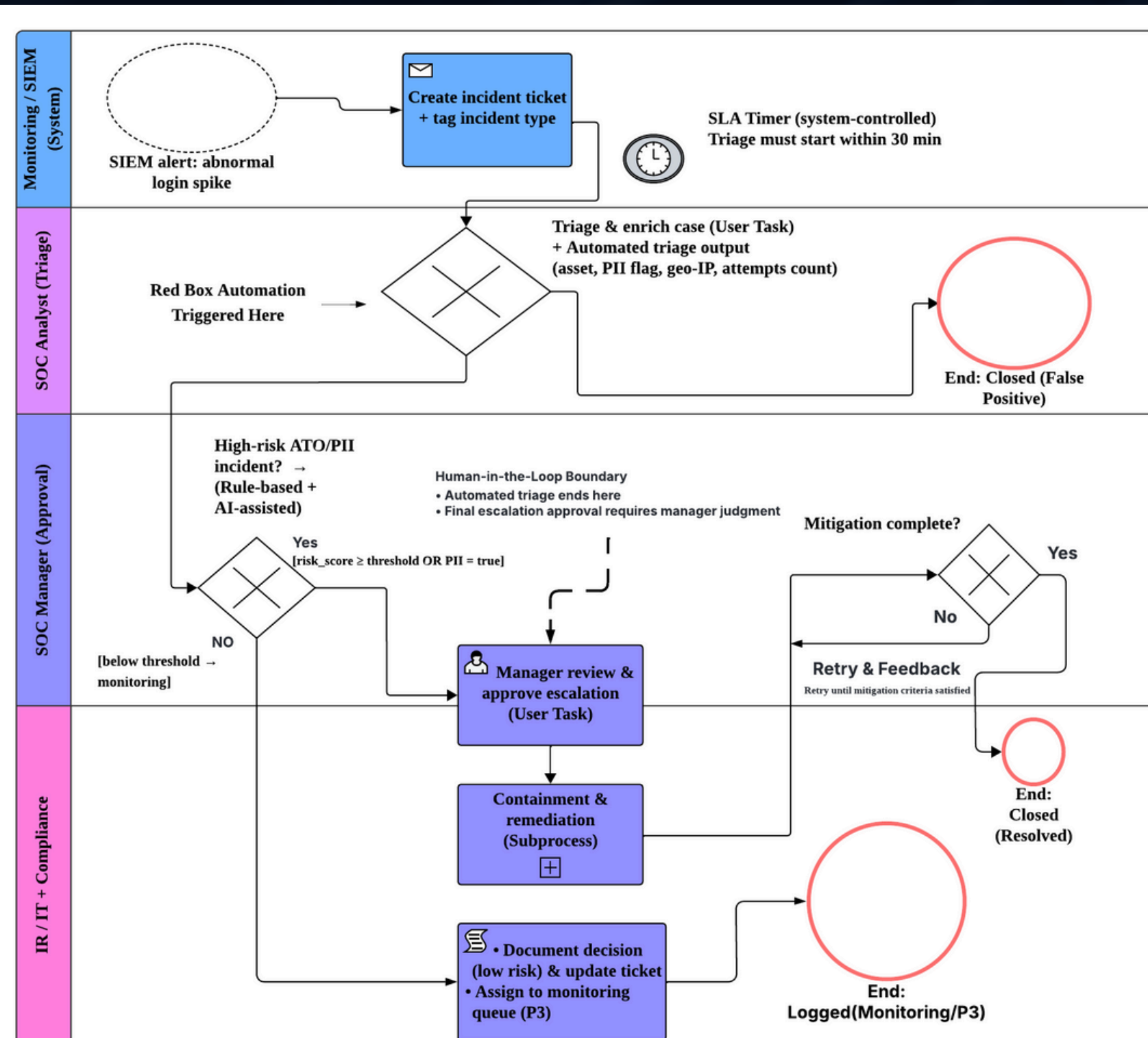


Color Key:
Red = High Intensity (High threat/power, bad for incumbents)
Yellow = Medium Intensity
Green = Low Intensity (Low threat/power, good for incumbents)
Dark Red = VERY HIGH Intensity (extreme pressure)

- Rivalry and buyer expectations rise → speed + auditability matter
- “Good-enough” automated alternatives increase pressure
- New vendor dependency (LLM/API) → validation and cost control needed
- Result: LLM-enabled workflow standardizes triage and accelerates escalation



BPMN Main Flow



Defines end-to-end roles: System → Analyst → Manager → IR/Compliance

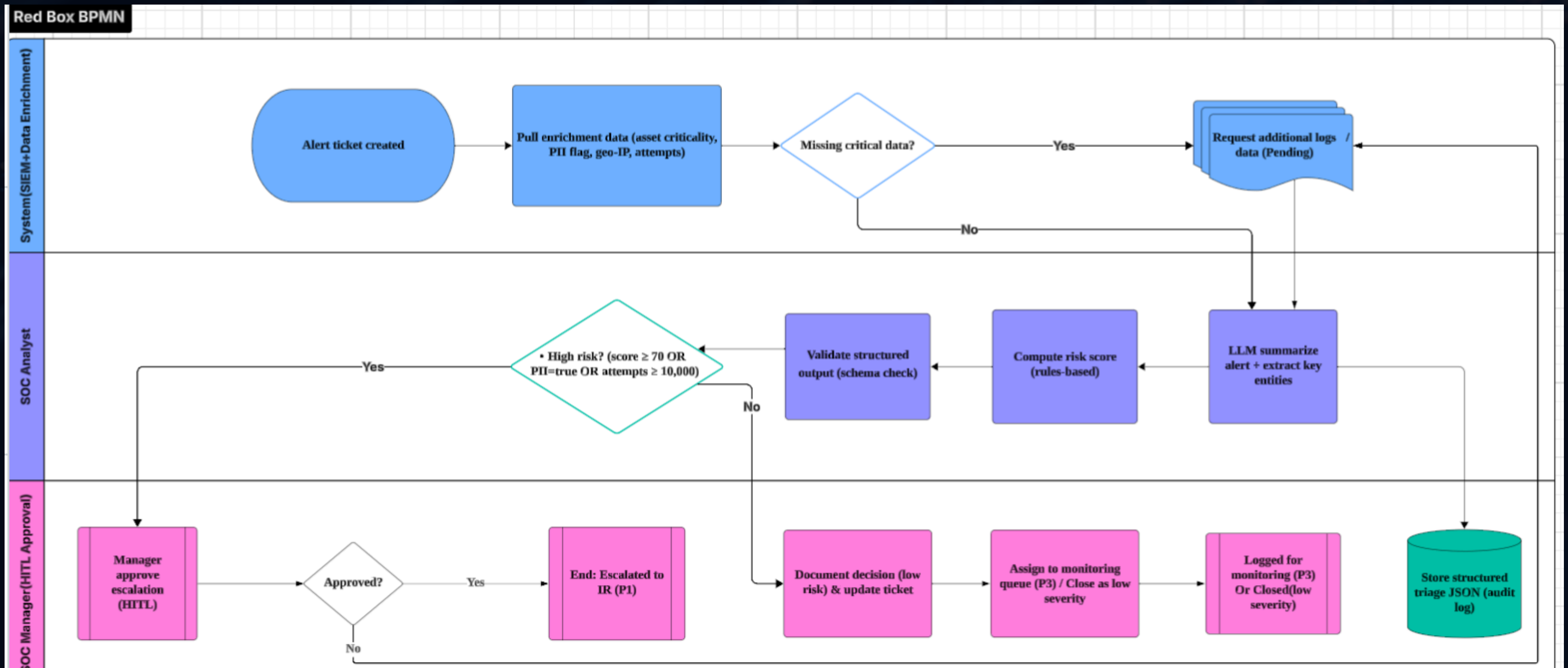
Key decisions: high-risk escalation vs low-risk monitoring/closure

Sets the control points for SLA, approval (HITL), and documentation

This is the operational blueprint: ownership and decision boundaries.

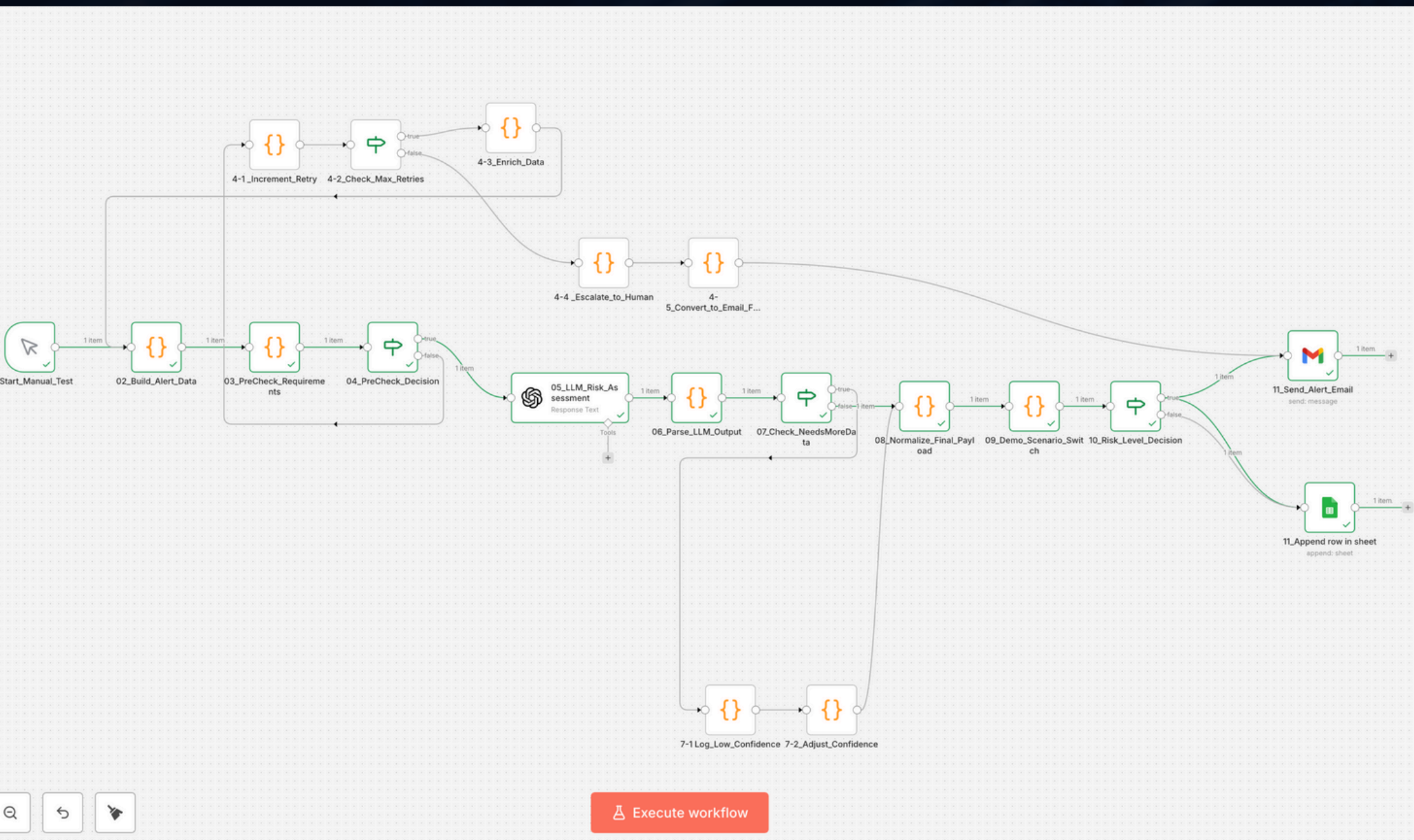


BPMN Red Box





Technology Stack — n8n



What n8n does in our system

- Runs the full pipeline: Input → LLM → Parse/Normalize → Route → Notify/Log
- Turns BPMN gateways into IF/ELSE routing
- Shows real-time execution for debugging + live demo

Key nodes we used

- Trigger: 01_Start_Manual_Test
- Build data: 02_Build_Alert_Data(Python)
- Pre-check: 03_PreCheck_Requirements → 04_PreCheck_Decision
- LLM call: 05_LLM_Risk_Assessment
- Code (JavaScript): 06_Parse_LLM_Output, 08_Normalize_Final_Payload
- Risk routing (IF): 10_Risk_Level_Decision
- Actions: 11_Send_Alert_Email, 11_Append_row_in_sheet
- Controls: Retry loop (4-1~4-3), Low-confidence (7-1~7-2), Demo switch (09)



BPMN → n8n “Executable Logic” Mapping

BPMN Key Step	n8n Node(1:1)	Core Output
Start / Alert input	01_Start_Manual_Test	Start workflow
Build incident payload	02_Build_Alert_Data	Structured alert JSON Creation
Pre-check (missing data?)	03_PreCheck_Requirements → 04_PreCheck_Decision	Determining whether missing data is/ quartering
LLM Analysis (summary + type + risk)	05_LLM_Risk_Assessment	risk level/score/type + summary/entities
Parse & Normalize output	06_Parse_LLM_Output → 08_Normalize_Final_Payload	clean JSON / Field organization
Risk Level Gateway (High risk?)	10_Risk_Level_Decision	IF routing (escalate vs log)
Escalation action	11_Send_Alert_Email	High Risk notification mail
Audit Log (all cases)	11_Append_row_in_sheet	Google Sheets recording(Audit tracking)



Live Showcasing

https://sejinkim.app.n8n.cloud/workflow/cO2wOMG_K-_GppOphxKT



Conclusion & Summary

7.1 Key Outcomes + Market Fit (B2B)

Key outcomes

- Speed: Automated triage reduces manual review time and accelerates escalation for high-risk cases
- Consistency: Structured LLM output + rule-based routing produces repeatable severity decisions
- Auditability: Every run is logged to Google Sheets (decision trail for reviews and compliance)

Expected competitors / platforms (examples)

- SOAR & automation: Palo Alto Cortex XSOAR, Splunk SOAR, Microsoft Sentinel (automation/playbooks)
- SIEM + workflow ecosystems: Splunk (SIEM), IBM QRadar, Google Chronicle
- Alerting / incident workflow: ServiceNow Security Operations, Jira + incident tooling
- (Our positioning: lightweight, customizable orchestration + LLM layer for triage and summaries)

Target B2B customers

- Telecom & ISPs (subscriber data, high-volume alerts)
- Large enterprises with SOC teams (finance, e-commerce, platform companies)
- Managed Security Service Providers (MSSPs) (multi-client triage efficiency)
- Regulated industries (healthcare, public sector, any org with strict audit/compliance needs)

THANK YOU

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